



Mo Tu We Th Fr Sa Su

Memo No.

Date

#20 Given $F_1 = (300.0\hat{i} + 500.0\hat{j})\text{ N}$, $F_2 = -200.0\hat{i}\text{ N}$
 $F_3 = -800.0\hat{j}\text{ N}$

(a) Net force in component form F_{net}
 $F_{\text{net}} = F_1 + F_2 + F_3$

• $F_x = 300.0 + (-200.0) + 0 = 100.0\text{ N}$

• $F_y = 500.0 + 0 + (-800.0) = -300.0\text{ N}$

$F_{\text{net}} = (100.0\hat{i} - 300.0\hat{j})\text{ N}$

(b) Magnitude and direction

• $|F_{\text{net}}| = \sqrt{F_x^2 + F_y^2} = \sqrt{(100.0)^2 + (-300.0)^2}$
 $= \sqrt{100000} = 316\text{ N}$

• $\theta = \tan^{-1}(F_y/F_x) = \tan^{-1}(-300.0/100.0) =$

$= \tan^{-1}(-3.00) = -71.6^\circ$

$F_x > 0$ and $F_y < 0$, so the force is in quadrant IV: 71.6° below the $+x$ axis.

#22 $F = 75.0\sqrt{2}(\hat{i} - \hat{j})\text{N}$, $F_2 = 150.0\sqrt{2}(\hat{i} - \hat{j})\text{N}$

Equilibrium Condition: $\Sigma F = 0$, so $F_1 + F_2 + F_3 = 0$
 therefore: $F_3 = -(F_1 + F_2)$

Sum: $F_1 + F_2 = (75 + 150)\sqrt{2}(\hat{i} - \hat{j}) = 225\sqrt{2}(\hat{i} - \hat{j})\text{N}$
 $N = (318.2\hat{i} - 318.2\hat{j})\text{N}$

Result $F_3 = -225\sqrt{2}(\hat{i} - \hat{j})\text{N} =$
 $\underline{\underline{(-318\hat{i} + 318\hat{j})\text{N}}}$



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#24 Given: $m = 63.0 \text{ Kg}$, $a = 4.200 \text{ m/s}^2$

2nd Newton law $F_{\text{net}} = m \cdot a$

$$F_{\text{net}} = (63.0 \text{ Kg})(4.200 \text{ m/s}^2) = \underline{\underline{264.600 \text{ N}}}$$

#34 $m = 1000 \text{ Kg}$, $v_0 = 0$, $v = 90.0 \text{ Km/h}$, $t = 10.0 \text{ s}$

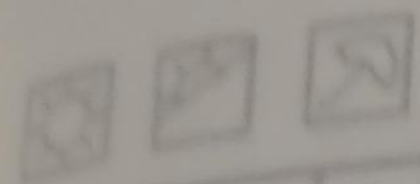
Unit conversion $v = 90 \text{ Km/h} \times (1000 \text{ m/1 Km}) \times$
 $(1 \text{ h/3600 s}) = 25.0 \text{ m/s}$

a) Acceleration $a = (v - v_0)/t = (25.0 - 0)/10.0 = \underline{\underline{2.50 \text{ m/s}^2}}$

b) net force $= F_{\text{net}} = m \cdot a = (1000.0 \text{ Kg})(2.50 \text{ m/s}^2)$
 $= \underline{\underline{2.50 \times 10^3 \text{ N}}}$

#54 $m_{\text{car}} = 550 \text{ Kg}$, $m_{\text{truck}} = 2200 \text{ Kg}$
 $a_{\text{truck}} = 10 \text{ m/s}^2$

3rd Newton law $[F_{\text{car on truck}}] = [F_{\text{truck on car}}]$



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$$\text{Force on the truck} = F = m \cdot a = (2200 \text{ kg})(10 \text{ m/s}^2) \\ = 2.2 \times 10^4 \text{ N}$$

$$\text{Car acceleration: } a = F/m = (2.2 \times 10^4 \text{ N}) / (550 \text{ kg}) \\ = \underline{40 \text{ m/s}^2}$$

$$\#108 \quad m = 1.67 \times 10^{-27} \text{ kg}, v_0 = 2.00 \times 10^5 \text{ m/s}, \\ v = 9.00 \times 10^5 \text{ m/s}, \Delta x = 10 \text{ cm} = 0.1000 \text{ m}, \\ \text{constant acceleration}$$

$$a = (v^2 - v_0^2) / (2 \Delta x)$$

$$a = [(9.00 \times 10^5)^2 - (2.00 \times 10^5)^2] / [2(0.100)]$$

$$a = (8.10 \times 10^{11} - 4.00 \times 10^{10}) / 0.200$$

$$a = (7.70 \times 10^{11}) / 0.200 = 3.85 \times 10^{12} \text{ m/s}^2$$

$$\text{Newton's 2nd law: } F = m \cdot a = (1.67 \times 10^{-27} \text{ kg}) \times \\ (3.85 \times 10^{12} \text{ m/s}^2) = 6.43 \times 10^{-15} \text{ N}$$

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$F = 30.0 \text{ N}$ for each force
 $M = 10.0 \text{ kg}$ frictionless surface

- One force F acts horizontally to the right
(+x)
- The other force pushes on the block
at -30.0° from the horizontal

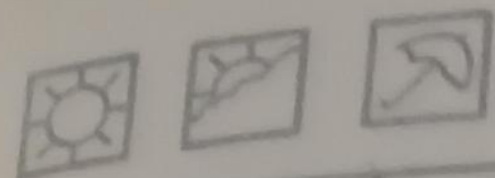
force components

$$F_1 = (30.0 \text{ N}) \hat{i}$$

$$F_2 = F \cos 30.0^\circ \hat{i} - F \sin 30.0^\circ \hat{j} = (25.98 \hat{i} - 15.0 \hat{j}) \text{ N}$$

$$y\text{-axis } N - Mg - F \sin 30.0^\circ = 0 \rightarrow N$$
$$\rightarrow (10.0)(9.80) + 15.0 = 113 \text{ N}$$

Since $N > 0$, the block stays in contact
with the surface, so all of the acceleration
is horizontal



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$$\text{x axis } \Sigma F_x = F + F \cos 30.0^\circ = 30.0 \text{ N} + 25.98 \text{ N} = 55.98 \text{ N}$$

$$a = \Sigma F_x / m = 55.98 \text{ N} / 10. \text{ K} = 5.598 \text{ m/s}^2$$

$$a \approx 5.60 \text{ m/s}^2$$